Project Synopsis: Multilingual Document Intelligence RAG

[PROJECT SYNOPSIS]
Title: Resource-Efficient, Citation-Grounded Multilingual Document Intelligence for Private Enterprise Archives

[ABSTRACT]
Organizations hold private documents in English and regional languages across PDFs, DOCX files, and scanned images. Keyword search misses semantically related passages, while general-purpose language models can provide unsupported answers. This project develops and evaluates a multilingual Retrieval-Augmented Generation system that retrieves evidence from private document libraries and produces answers with page-level citations. The central study compares direct multilingual embeddings, translation-to-English embeddings, and an adaptive hybrid retrieval route. The system combines OCR-aware ingestion, semantic and lexical search, cross-encoder reranking, a lightweight answer generator, and abstention when evidence is insufficient. The intended contribution is a reproducible evaluation of retrieval accuracy, citation faithfulness, language fidelity, OCR robustness, latency, and resource cost for English and selected Indian languages.

[PROBLEM STATEMENT]
Existing multilingual RAG research usually concentrates on one dimension: translation, a single low-resource language, multimodal retrieval, or a narrow legal corpus. There remains insufficient evidence on how to build a trustworthy and resource-efficient system for heterogeneous private documents in English and Indian languages, especially when source citations, OCR noise, language consistency, and constrained hardware are important.

[AIM]
To design and evaluate a citation-grounded multilingual RAG system for private document collections that selects or combines retrieval strategies according to language, query, and document characteristics.

[OBJECTIVES]
1. Ingest native PDFs, DOCX files, and scanned documents through parsing, OCR, language identification, and page-aware chunking.
2. Compare direct multilingual embedding, translation-based embedding, and hybrid dense-plus-BM25 retrieval.
3. Improve ranking using a multilingual cross-encoder reranker.
4. Generate concise answers with page-level document citations and an explicit insufficient-evidence response.
5. Evaluate retrieval quality, faithfulness, answer-language fidelity, latency, and resource use for English and selected Indian languages.

[PROPOSED METHODOLOGY]
Ingestion: Extract text and metadata from PDF/DOCX; use Tesseract OCR for scans; detect language; split text into overlapping, page-preserving chunks.
Indexing: Store chunk text, language, document ID, page number, and embeddings in PostgreSQL with pgvector HNSW. Maintain BM25 lexical retrieval alongside dense retrieval.
Retrieval experiment: Run three conditions: (A) multilingual-E5 direct retrieval, (B) query/document translation followed by English embedding, and (C) hybrid dense-plus-BM25 retrieval with cross-encoder reranking. An adaptive router may select the route based on

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language confidence and query features.
Generation and trust: Give the generator only reranked evidence chunks. Return cited claims with document/page references. If retrieval confidence or evidence coverage is low, return an abstention with suggested query refinement.
Delivery: FastAPI backend with asynchronous ingestion; a Next.js interface for upload, querying, citation viewing, and streamed answers. Workspace isolation is an engineering feature, not the central research claim.

[EVALUATION PLAN]
Dataset: Build a consented/private or public evaluation collection containing English and selected Indian-language documents, native PDFs and OCR scans, with manually checked question-answer-evidence triples.
Retrieval metrics: Recall@k, MRR@10, nDCG@10, and per-language results.
Answer metrics: human correctness, citation precision/recall, faithfulness or groundedness, and abstention precision for unanswerable questions.
Operational metrics: end-to-end latency, throughput, GPU/CPU memory use, and cost per query.
Robustness metrics: performance by document format, OCR quality, cross-lingual query-document pair, and requested answer language.
Baselines: vanilla dense multilingual RAG, translation-only RAG, and hybrid retrieval without reranking or abstention.

[EXPECTED CONTRIBUTIONS]
1. A controlled comparison of multilingual, translation-based, and hybrid retrieval for English and Indian-language private documents.
2. A page-citation and evidence-sufficiency mechanism for more trustworthy answers.
3. An evaluation protocol covering retrieval, groundedness, language behaviour, OCR robustness, latency, and cost.
4. A deployable open-source-oriented architecture suitable for constrained academic hardware.

[SCOPE AND LIMITATIONS]
The first implementation should claim support only for languages actually evaluated, for example English, Hindi, Marathi, and one or two additional Indian languages. Table/chart understanding requires a dedicated vision-language component and should be presented as future work unless implemented and tested. The system assists document search and does not provide legal, medical, or compliance advice.

[KEY REFERENCES]
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